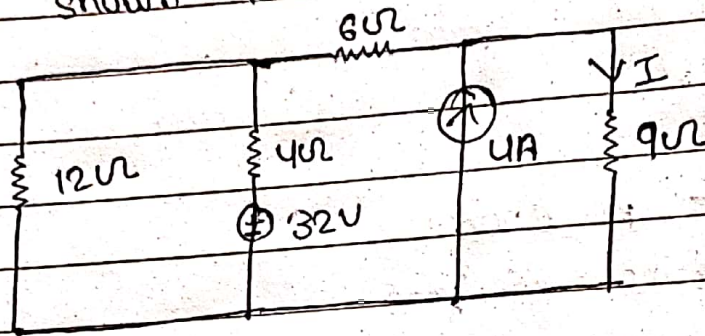
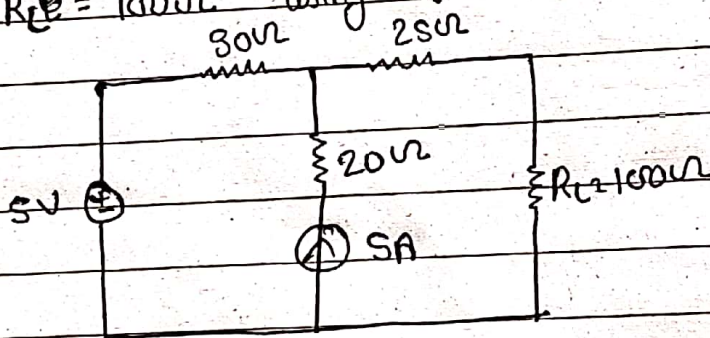


## Important Question BEE

1. Explain the nodal analysis method with the help of example. or
2. Explain the mesh analysis method with the help of example.
3. State superposition theorem and example explain with the help of example. or
4. State Thevenin theorem and explain with the help of example.
5. Determine the power dissipated in  $9\Omega$  resistance of circuit shown in below— using by superposition theorem



6. In the circuit shown in figure find the value of current in  $R_L = 100\Omega$  using Thevenin theorem



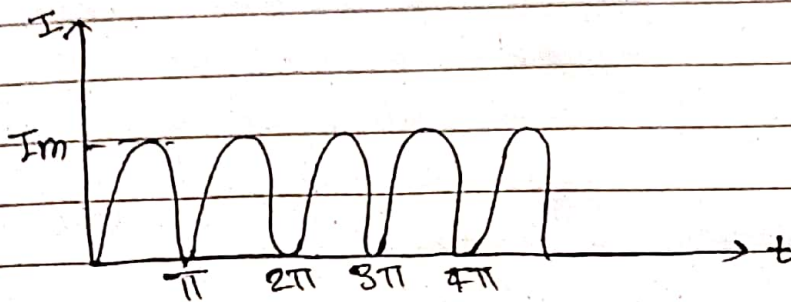
Define avg. value, effective value, form factor of an alternative current.

Show that the current through purely resistive and purely inductive circuit is in phase with the applied

Voltage. or

Show that the current through R-L-C circuit is in phase with the applied voltage.

Ques Find the average value, rms value, form factor and peak factor of the current shown in figure



Ques What does exactly mean of semiconductor. Explain types of semiconductor with the help of crystalline structure at  $0K$  and  $300K$ .

Ques Explain the p-n junction diode. And explain working principle of p-n junction diode. or

Explain the working principle of p-n junction diode when it's forward bias.

Ques Why we called BJT <sup>work</sup> as a amplifier? or

Explain working principle of NPN transistor with help of diagram. or

Explain the characteristic of C.E. and C.B ~~config~~ of BJT with the help of input and output characteristic

Ques Explain the SHR with the help of block diagram.



Que 10 Comparison b/w A.M and F.M ? atleast 10 points  
Que 11 explain frequency modulation with the help of diagram  
(derivation of f.m)

Que 12 write the short note on satellite communication /  
multiplexing with the help of diagram.

Que 13 explain the working principle of following

- (i) UJT
- (ii) SCR
- (iii) TRIAC